Report Task 5

Code Added

I added this code in the Inference section after the trace plots , it generates a tabular summary of the posterior distributions and extract the associated R(hat) values:

```
az.plot_trace(
idata, var_names=["~mu", "~month mu_raw"])
summary = az.summary(
    idata,
    var_names=["intercept", "linear trend",
               "temp_coeff", "sigma",
               "month_mu"],
    filter_vars="like"
)
print(summary)
summary.to_csv(
"posterior_summary_excess_deaths.csv")
```

Reported R(hat) Values

All parameters show R(hat) values between **1.00 and 1.01**. This includes the intercept, linear trend, temperature coefficient, noise scale, and all monthly effects. These values indicate excellent chain convergence and no evidence of poor mixing or divergent chain behaviour.

Table 1: Posterior summary (pre < 2020)

Parameter	Mean	SD	R(hat)
intercept	44555.112	1111.273	1.01
temp coeff	-423.167	113.662	1.01
linear trend	22.880	4.211	1.00
sigma	2659.505	151.820	1.00
month mu[January]	7535.910	884.215	1.00
month mu[February]	-577.531	882.857	1.00
month mu[March]	1436.586	764.311	1.00
month mu[April]	-88.284	651.796	1.00
month mu[May]	-812.904	676.375	1.00
month mu[June]	-1269.456	805.401	1.00
month mu[July]	-536.766	947.522	1.00
month mu[August]	-2288.643	921.971	1.00
month mu[September]	-2279.057	791.220	1.00
month mu[October]	-307.837	687.536	1.00
month mu[November]	-1460.016	737.038	1.00
month mu[December]	647.999	852.998	1.00

Why R(hat) Is Important

The R(hat) compares between-chain and within-chain variance to assess convergence. Values close to 1 imply that all MCMC chains are sampling from the same posterior region. If R(hat) substantially exceeds 1, the sampler has not converged and posterior estimates may not be very reliable.

Counterfactual Prediction Error

It is unable to represent the large structural break introduced by COVID-19. Since it was trained only on pre-COVID data, it assumes that pre-pandemic patterns persist unchanged and COVID-19 is a very unique time period. When mortality increases dramatically, the model's baseline predictions stay close to old levels so it underestimates the number of deaths.

Report Task 6

Data split and model fitting

```
pre = df[df.index < "2010"]
post = df[df.index >= "2010"]
```

I refit the same PyMC model on `pre` and summarised posteriors with `az.summary`, reporting mean, SD and $\hat{R}$.

Comparison with Original Pre-2020 Posteriors

I refit the same PyMC model on `pre` and summarised posteriors. Using pre-2020 data gives a clearer, tighter positive linear trend and narrower posteriors for seasonal effects. Restricting to pre-2010 reduces data, so intervals widen and trend certainty decreases; temperature and seasonality remain broadly similar.

Table 2: Posterior summary (pre < 2010, key parameters)

Parameter	Mean	SD	R(hat)
intercept	45159.23	1394.46	1.00
temp coeff	-392.79	139.76	1.00
linear trend	13.06	11.05	1.00
sigma	2580.25	312.08	1.00

Report Task 7

In the full model, predictions follow a clear pattern higher deaths in winter and lower in summer. without months indicator the model is still similar, because the effects of temperature remain although peaks shrink. When both month and temperature are removed, predictions are almost flat with only a rising mean and wide intervals; spikes vanish because seasonality is unaccounted for.

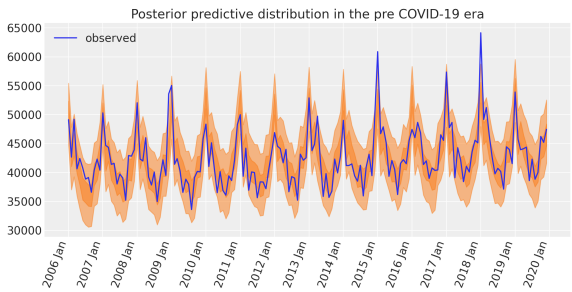

Figure 1: Posterior predictive distribution in the pre COVID-19 era (full model).

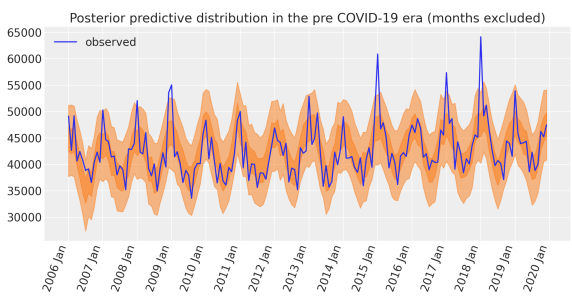

Figure 2: Posterior predictive distribution in the pre COVID-19 era (month indicators excluded).

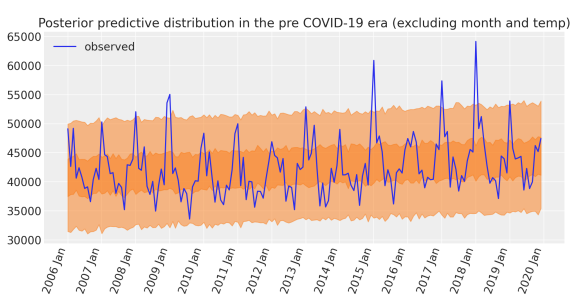

Figure 3: Posterior predictive distribution in the pre COVID-19 era (month and temp excluded).

Report Task 8

Using the sequences I generated, I plotted an error measure that shows how accurate the model samples the category and plots that compare each model to real deaths. Both models get the overall category proportions roughly correctly($\approx (0.335, 0.330, 0.335)$, HMM1 $\approx (0.339, 0.311, 0.350)$, HMM2 $\approx (0.331, 0.336, 0.333)$ (Table 3), and HMM2 is closest numerically to the overall proportion of deaths, however, HMM1 produces longer, smoother runs of highs and lows and lines up with the real series more often month by month, so it looks more realistic overall. Differences are noticeable but not extremely significant.

Table 3: Empirical frequencies of death categories (low, medium, high).

Model	Low	Medium	High
Real	0.335	0.330	0.335
HMM1	0.339	0.311	0.350
HMM2	0.331	0.336	0.333

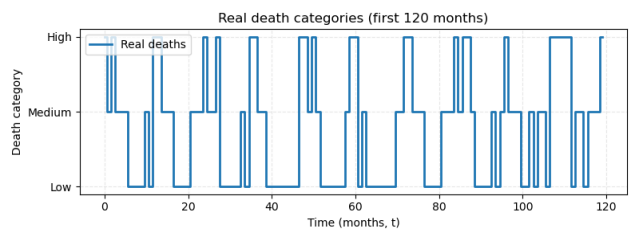

Figure 4: Real vs HMM-generated death–category sequences (first 120 months).

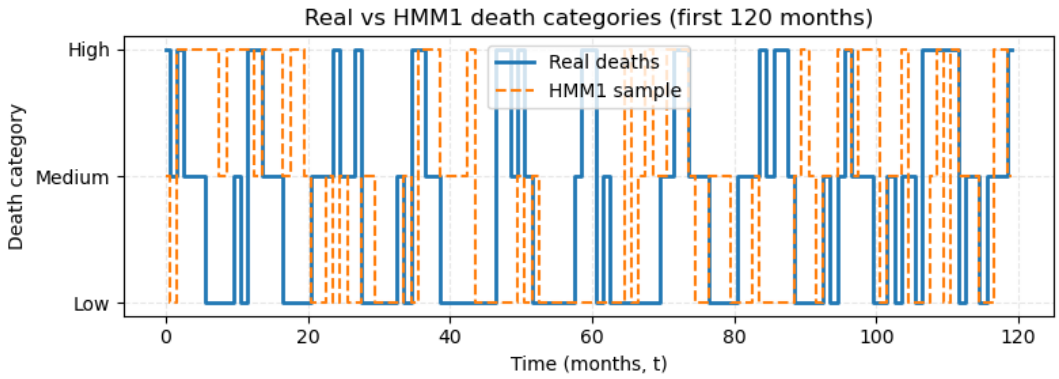

Figure 5: Real vs HMM-generated death–category sequences (first 120 months).

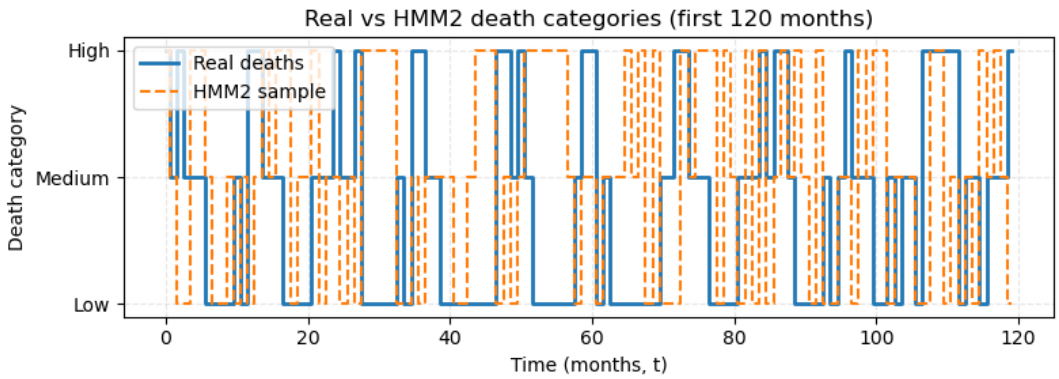

Figure 6: Real vs HMM-generated death–category sequences (first 120 months).

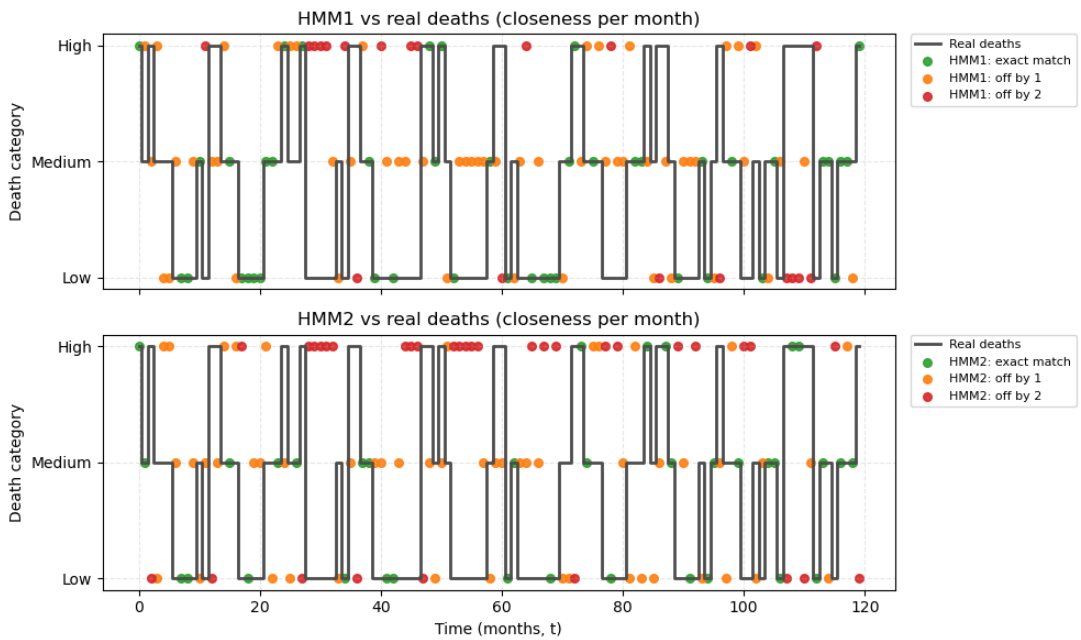

Figure 7: Real vs HMM-generated death–category sequences (first 120 months).